

Internship Plan

INTERNSHIP PLAN BY DATAROPES.AI
DATA ENGINEERING

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Data Engineering Internship Plan

Duration: 12–14 Weeks

Level: Beginner → Job-Ready Junior Data Engineer

MONTH 1 — DATA & SQL FOUNDATIONS

◇ Week 1: SQL Fundamentals (Mandatory)

Topics:

- SELECT, WHERE, ORDER BY
- JOIN (INNER, LEFT, RIGHT)
- GROUP BY, HAVING
- COUNT, SUM, AVG
- CASE WHEN

Hands-on:

- Query customer & order tables
- Calculate revenue per customer
- Filter by date ranges

Free YouTube References

- *SQL Full Course (Beginner)* – freeCodeCamp
<https://www.youtube.com/watch?v=HXV3zeQKqGY>
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◇ Week 2: Advanced SQL I (Analytics)

Topics

- **Window Functions**
 - ROW_NUMBER() – deduplication
 - RANK() / DENSE_RANK() – top-N problems
 - LAG() / LEAD() – growth & churn
- **CTEs**
 - Breaking complex logic into readable steps
- **Conditional Aggregation**
 - SUM(CASE WHEN ... THEN ... END)

Hands-on

- Monthly revenue growth
- Top products per category
- Running totals

Free YouTube References

- *Advanced SQL – Window Functions*
<https://www.youtube.com/watch?v=Ww71knvhQ-s>
- *CTEs Explained*
<https://www.youtube.com/watch?v=QNfnuK-1YYY>

◇ Week 4: Python + SQL Integration

Topics:

- Running SQL from Python
- Parameterized queries
- Bulk inserts
- Error handling & retries

Hands-on

- Python script:
 - Read CSV
 - Transform
 - Insert into database

Free YouTube References

- *Python + SQL Tutorial*
<https://www.youtube.com/watch?v=uL0-6kfiH3g>

MONTH 2 — ADVANCED SQL & DATA ENGINEERING LOGIC

◆ Week 5: Data Engineering Fundamentals

Topics:

- **What is Data Engineering**
 - Building pipelines that move data from source → storage → analytics
- **OLTP vs OLAP**
 - OLTP: transactional systems (Shopify, Stripe)
 - OLAP: analytical systems (BigQuery, Snowflake)
- **Batch vs Streaming**
 - Batch: scheduled loads
 - Streaming: real-time data
- **High-level Architecture**
 - Source → Ingestion → Storage → Transformation → BI

Hands-on:

- Draw pipeline architecture diagram
- Install Python, Git, VS Code

Free YouTube References

- *What is Data Engineering?*
<https://www.youtube.com/watch?v=qWru-b6m030>
- *OLTP vs OLAP Explained*
<https://www.youtube.com/watch?v=ksDfqj9045Y&pp=ygUUb2x0cCB2cyBvbGFwIGVkdXJla2HSBwkJTTwoBhyohjO8%3D>

◇ Week 6: Incremental Loads & Deduplication

Topics:

- **Incremental loading**
 - Load only new/updated data
- **Deduplication**
 - Use ROW_NUMBER() over partition
- **SCD Type 1 & 2**
 - Track historical changes

Hands-on

- Build incremental pipeline using timestamp
- Deduplicate incoming data

Free YouTube References

- *Slowly Changing Dimensions Explained*
<https://www.youtube.com/watch?v=Fbp3j5m5g5k>

◇ Week 7: Advanced SQL II (Business Logic)

Topics:

- **Cohort Analysis**
 - Group users by acquisition month
- **Churn Calculation**
 - Identify churned users over time
- **Funnel Analysis**
 - Step-by-step drop-off tracking

Hands-on

- Subscription churn query
- Retention cohort matrix

Free YouTube References

- *Cohort Analysis in SQL*
<https://www.youtube.com/watch?v=LXqpx9mr0Is>
-

◇ Week 8: SQL Performance Optimization

Topics:

- Indexes
- Partitioning & clustering
- Query execution plans
- Cost optimization

Hands-on

- Optimize slow queries
- Compare before vs after execution time

Free YouTube References

- *SQL Performance Tuning*
https://www.youtube.com/watch?v=-t-8-xoLyv4&list=PLf-39075ydszFgJWvnESuz_u_pemT9eie
-

◇ Week 9: SQL for Data Warehouses

Topics

- Nested data (ARRAY, STRUCT)
- UNNEST
- JSON extraction
- Dynamic SQL (intro)

Hands-on

- Flatten nested JSON
- Query API data stored in warehouse

Free YouTube References

- *BigQuery Nested & Repeated Fields*
<https://www.youtube.com/watch?v=KHjJxZFUNHQ>
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If you finish the above within one month (optional)

Topics:

- DAGs
- Dependencies
- Idempotency
- Retry logic

Hands-on

- Build Airflow DAG
- Schedule SQL pipeline

Free YouTube References

-  *Apache Airflow Full Course*
https://www.youtube.com/watch?v=K9AnJ9_ZAXE
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Topics:

- Staging → Intermediate → Mart layers
- dbt-style transformations
- SQL testing

Hands-on:

- Build fact & dimension tables
- Apply business logic in SQL

Free YouTube References

-  *dbt Fundamentals*
<https://www.youtube.com/watch?v=5rNquRnNb4E>
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Topics:

- KPI definitions
- Semantic layer
- Optimized queries for dashboards

Hands-on

- Create Looker / Power BI dashboard
- Build SQL views

Free YouTube References

- *Looker Studio Tutorial*
<https://www.youtube.com/watch?v=K4Fh6dWqXkI>
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Project Requirements

- API + CSV ingestion
- Cloud storage
- Incremental SQL pipelines
- Advanced SQL analytics
- Dashboard

Deliverables

- GitHub repo
 - SQL folder structure
 - Architecture diagram
 - Final presentation
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Recommended SQL Project Structure

sql/

|— staging/

|— intermediate/

|— marts/

|— incremental/

|— tests/

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